Is the 360° street view redundant?

Phase-2 attention pruning — Qwen3.6-27B, $k=1$, $n=1663$ solved questions

The question

Each urban VQA question shows **4 street-view cameras** (`fwd`, `bwd`, `cross_left`, `cross_right`).

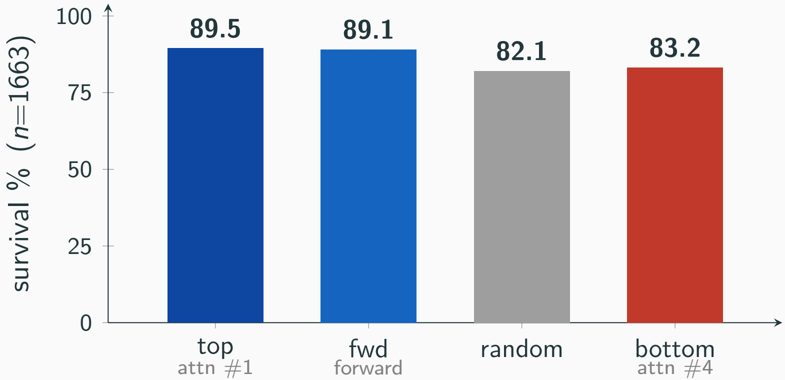
Hypothesis

The full 360° view is **redundant**: the **one image the model attends to most** keeps the answer correct — and that image is the **forward camera**.

Method. Solve each question on all 4 images → keep only the single chosen image → re-ask greedily.
Survival % = fraction of solved questions still correct on the kept image.

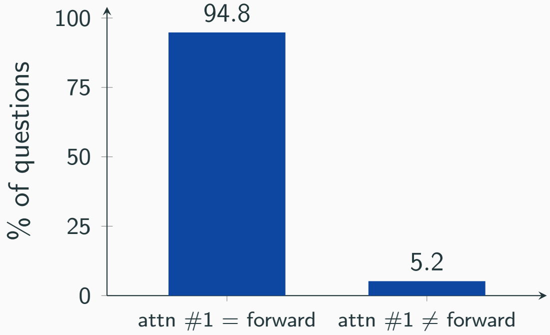
Arms: **top** (most-attended) | **fwd** (always forward) | random | **bottom** (least-attended).

One image is enough



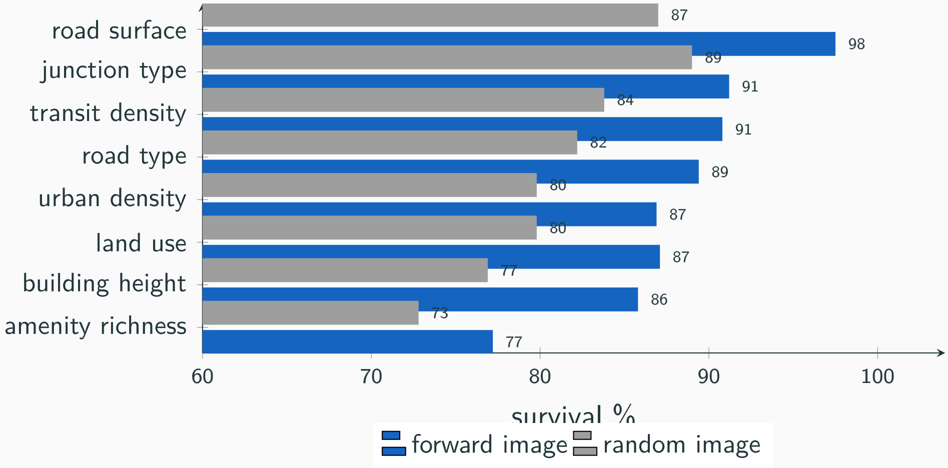
top = **fwd** ≫ random. The single most-attended image keeps the answer correct **9 times in 10**.

The attended image is the forward camera



- Attention's #1 image **is the forward camera in 94.8%** of questions.
- Pooled survival: top 89.5% ≈ fwd 89.1% — picking by attention and "always forward" land in the same place.
- Practical rule: **keep the forward image, drop the other three**.

Holds across every topic



In all 8 tasks the **forward image** beats a **random** one — the answer lives in the forward view everywhere.

Take-home

- ✓ The 360° view is **redundant** — **one** street-view image suffices (89.5% survival).
- ✓ That image is the **forward camera** (`along_fwd`), chosen by attention 94.8% of the time.
- ✓ The right image **beats a random one** by +7 pts, in **every** topic.

3× fewer image tokens per question, same answer.

Repro: `python attention_prune/analyze_prune_k1.py` | n=1663, sha `b083f41`.